



Exercises



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Exercises

- ▶ Consider an independent and periodic task set, executed in a pre-emptive context:

	C (ms)	T (ms)	D (ms)	Prio
τ_1	2	6	5	
τ_2	3	12	12	
τ_3	4	15	15	

- ▶ Assume a fixed priority assignment, based on Rate Monotonic. Assign priorities accordingly (fill in the table above): 3 (higher), 2 and 1 (lower). Justify your answer
- ▶ For the previous task set, represent the time diagram for the RM scheduling, considering the time interval that validates the schedulability of the task set

Exercises

- ▶ For the previous task set example, consider that the worst-case execution time of task 3 is 5 time units instead

	C (ms)	T (ms)	D (ms)	Prio
τ_1	2	6	5	
τ_2	3	12	12	
τ_3	5	15	15	

- ▶ Compute its worst-case response time

Exercises

- ▶ Consider the following taskset

	C (ms)	T = D (ms)	Arrival instant
τ_1	2	6	2
τ_2	3	12	1
τ_3	5	15	0

- ▶ Is this schedulable under preemptive EDF?
- ▶ Trace the execution of the taskset in the first 20 ms

Exercises

- ▶ Consider the following taskset

	C (ms)	T = D (ms)
τ_1	2	5
τ_2	1	9
τ_3	4	12

- ▶ Is this schedulable under non pre-emptive EDF?
- ▶ Trace the execution of the taskset in the first 20 ms

Exercises

▶ Exercise

- ▶ Consider two periodic tasks with computation times $C_1 = 1$, $C_2 = 2$ and periods $T_1 = 5$, $T_2 = 8$, handled by Rate Monotonic
- ▶ Show the schedule produced by a Sporadic Server, having maximum utilization and intermediate priority, on the following aperiodic jobs:
 - ▶ $\text{job}_i(a_i, C_i)$; $J_1 = (2, 3)$; $J_2 = (7, 1)$; $J_3 = (17, 1)$; where a_i = arrival time of job i

Exercises

► Exercise

- Consider two periodic tasks with computation times $C_1 = 1$, $C_2 = 1$ and periods $T_1 = 3$, $T_2 = 6$, handled by EDF
- Show the schedule produced by an additional CBS Server, having 25%, $(Q, T) = (1, 4)$, on the following aperiodic jobs:
 - $\text{job}_i (a_i, C_i)$; $J_1 = (2.2, 0.5)$; $J_2 = (7.5, 1)$; $J_3 = (10.2, 0.4)$, where a_i = arrival time of job i